TASK 4

Design for Testability (DFT) & Physical Verification

1. Aim

To understand the concepts of Design for Testability (DFT) and physical verification techniques such as DRC and LVS in VLSI design.

2. Objective

- To study the importance of testing in VLSI circuits
- To understand scan chain technique
- To learn about DRC (Design Rule Check) and LVS (Layout vs Schematic)

3. Tools Used

- Theoretical study
- Reference diagrams

4. Theory

- **4.1 Design for Testability (DFT)**
- Design for Testability is a technique used to make digital circuits easier to test after manufacturing. It helps in detecting faults and improving product reliability.

4.2 Need for DFT

- Detect manufacturing defects
- Improve test coverage
- Reduce testing time and cost

4.3 Scan Chain Concept

A scan chain is a technique where flip-flops are connected in a chain to form a shift register.

Working:

- In test mode, data is shifted through flip-flops
- Outputs are observed to detect faults

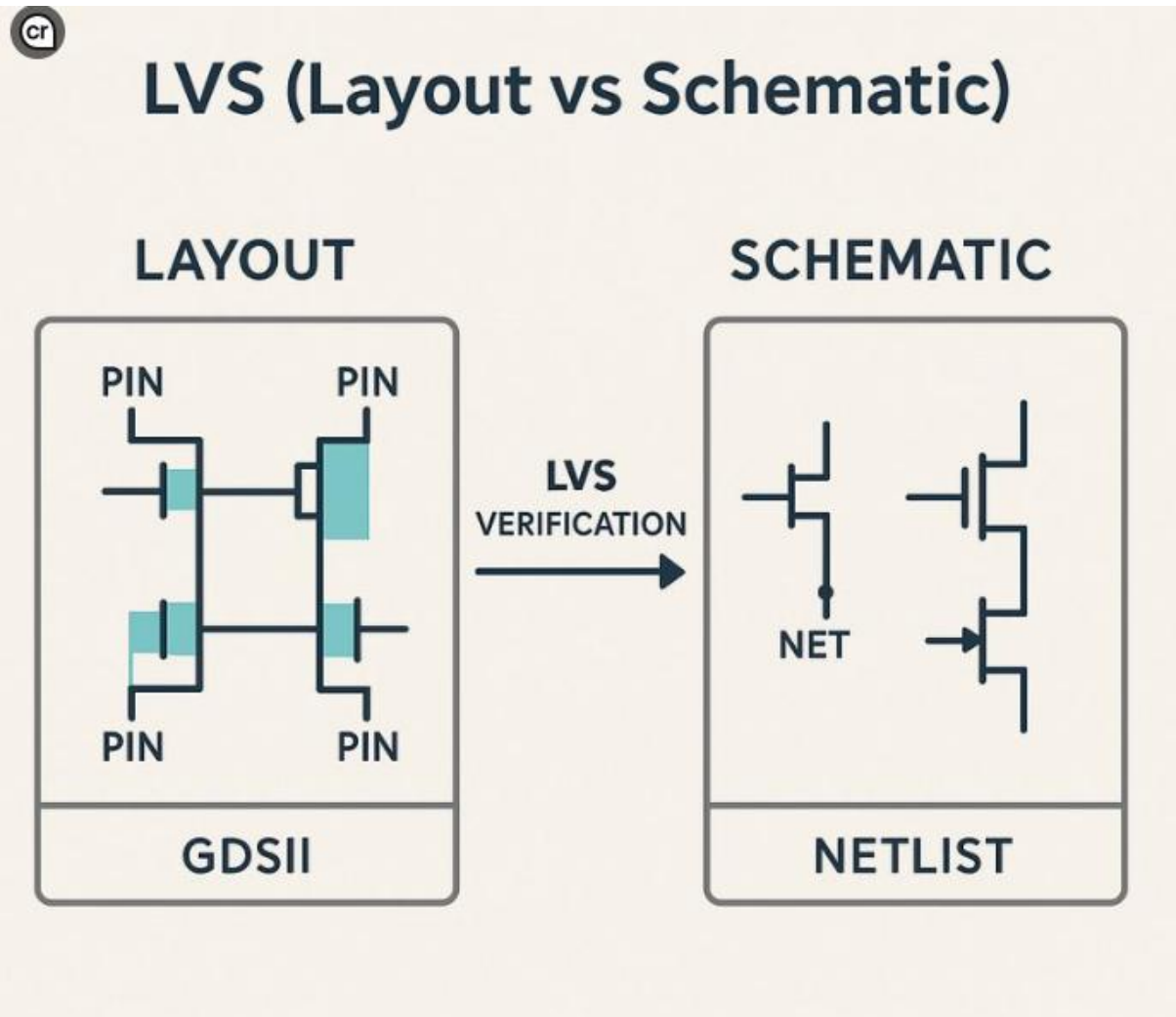
- **4.4 Physical Verification**
- Physical verification ensures that the fabricated layout matches the design rules and schematic.

4.5 DRC (Design Rule Check)

DRC checks whether the layout follows manufacturing rules such as:

- Minimum width
- Spacing between components
- Layer constraints

- **4.6 LVS (Layout vs Schematic)**



This diagram shows the comparison between the physical layout and schematic design using LVS. It ensures that both designs are electrically equivalent before fabrication.

5. Diagrams

1. Scan Chain

A scan chain links flip-flops in a circuit together into a shift register format during test mode. It converts internal state elements into controllable and observable points.

- **Structure:** Comprises Scan Input (SI), Scan Output (SO), and Scan Enable (SE) signals.
- **Process:** In "Shift" mode (SE=1), data is shifted through the chain. In "Capture" mode (SE=0), the circuit operates normally to capture functional data.

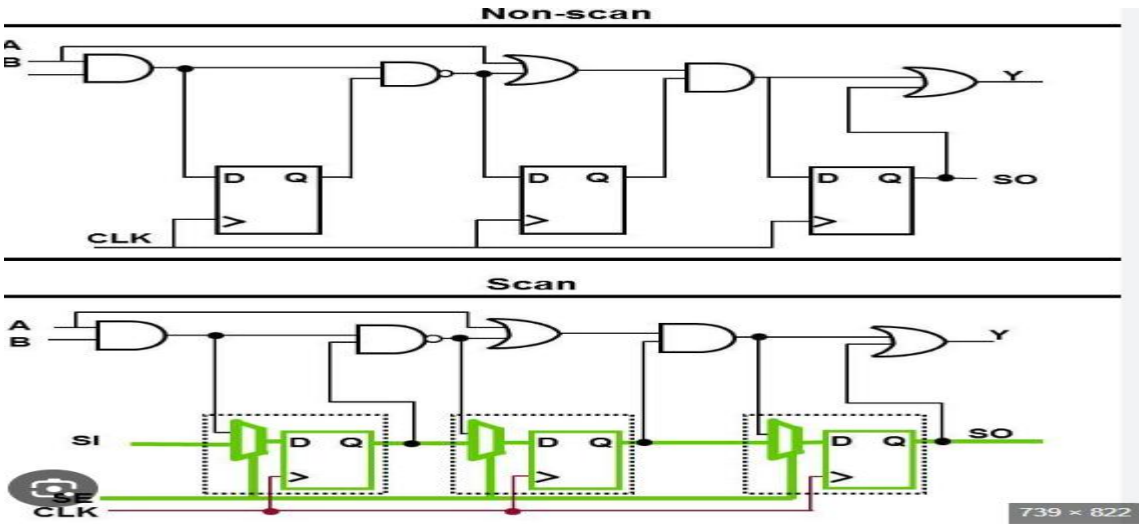


Figure 1: Comparison of a non-scan vs. scan chain (shift register) configuration, showing SI/SO connectivity.

2. DRC (Design Rule Check) Example

DRC ensures the physical layout is manufacturable and adheres to foundry rules (spacing, width, density).

- **Common Error:** Minimum Spacing Violation.
- **Example:** Two metal layers (Metal 1) are placed too close to each other, violating the minimum spacing rule.
- **Fix:** Increase the distance between the polygons to meet the foundry's minimum spacing requirement.

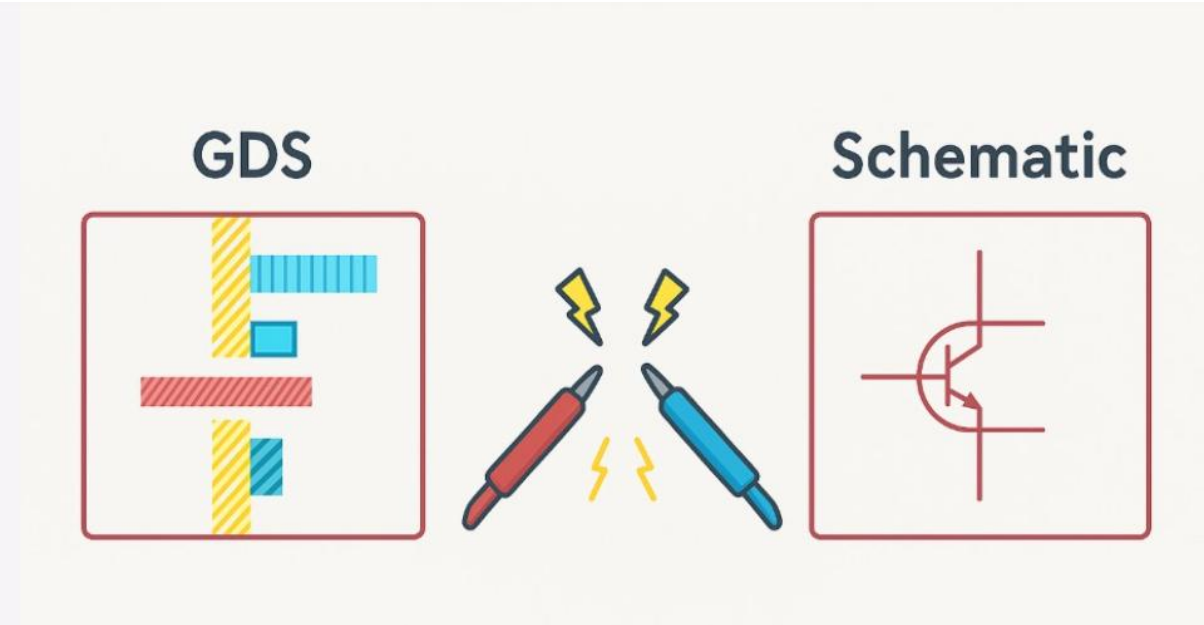


Figure 2: Example of a DRC violation where spacing between polygons is insufficient, and its corrected form.

6. Applications

- Used in chip manufacturing
- Ensures error-free IC production
- Improves reliability of electronic devices
-

7. Result

- The concepts of DFT, scan chains, DRC, and LVS were studied and understood successfully.

8. Conclusion

DFT techniques improve the testability of VLSI circuits, while DRC and LVS ensure correctness of layout design. These processes are essential for reliable chip fabrication.